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Review article

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Nitrous oxide sources, mechanisms and mitigation

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Abstract

Atmospheric nitrous oxide (N_2O) is a potent greenhouse gas and ozone-depleting substance. In this Review, we outline global N_2O sources, with a focus on hotspots and hot moments, and discuss strategies to mitigate N_2O emissions. N_2O can be released by natural sources such as bedrock weathering, but anthropogenic sources such as agriculture account for 40% of total emissions. Hotspots are localized regions of high emissions and include cropland soils (2.1 Tg N yr^{-1}), tropical forests (1.55 Tg N yr^{-1}), pasture soils with animal waste return (1.7 Tg N yr^{-1}), and streams and small lakes (0.4 Tg N yr^{-1}). Brief periods of intense emissions, known as hot moments, include post-deforestation, upland soils after fertilizer application, and desert and grasslands after precipitation. N_2O production from terrestrial and aquatic environments is mainly driven by two microbial processes: nitrification and denitrification. Bioaugmentation and biogeoengineering technologies hold potential for reducing N_2O emissions; for example, nature-based anammox hotspot geoengineering in Jiaxing, China, reduces N_2O emissions by 27.1%. However, the spatiotemporal heterogeneities and different production pathways of N_2O emissions are poorly represented in existing models, hindering the quantification and mitigation of emissions. A global N_2O database is needed to address this limitation. Additionally, artificial intelligence technology could enable real-time agricultural management to align nitrogen supply with crop demand.

Sections

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Introduction

Nitrous oxide (N_2O) contributed to 6.4% of the total warming caused by greenhouse gases during 1750–2022, ranking third after carbon dioxide (CO_2) and methane (CH_4)[1], and is the dominant stratospheric ozone-depleting substance[2]. Global temperature increases are projected to surpass 1.5 °C and even 2 °C by 2100 (ref. 3) even in scenarios with immediate fossil fuel cessation, highlighting the urgent need to mitigate N_2O emissions. The atmospheric concentration of N_2O increased from 270 ppb in 1850 to 338.3 ppb in 2024, and the average rate of increase has increased from 0.82 ± 0.33 ppb yr^{-1} in 1980–2020 to 1.15 ± 0.12 ppb yr^{-1} in 2021–2024 (refs. 4,5) (Fig. 1). Anthropogenic factors are a key driver of this growth, with sources including direct emissions from agriculture and industry, and indirect emissions from inland water, microplastics or aquatic nitrogen originating from agricultural fertilizers. Agricultural activities are responsible for over half of global anthropogenic N_2O emissions[5,6], creating an urgent need to reconcile food security with climate stabilization.

The production of N_2O from terrestrial and aquatic environments is mainly driven by two microbial processes: nitrification and denitrification. Nitrification is driven by autotrophic microorganisms and occurs under oxic conditions, whereas denitrification is driven by heterotrophic microorganisms and occurs under anoxic conditions. Although these two processes operate under very different environmental conditions, they are not differentiated in approaches such as the emission factor method, which is widely used for making inventories of N_2O emissions[7].

N_2O emissions show substantial spatiotemporal heterogeneity[5,6]. This variability manifests as spatially confined zones of disproportionately high emissions, known as hotspots, and brief periods of intense emissions, known as hot moments[8]. Hotspots occur from microsite to regional scales[9,10]. For example, the rate of N_2O release from fertilized croplands can be one to two orders of magnitude higher than from other dryland ecosystems. Hot moments are typically triggered by environmental conditions[11,12]. For instance, N_2O emissions from arid desert ecosystems are negligible during dry periods but can rapidly increase following rainfall events.

A comprehensive understanding of N_2O sources and mechanisms is urgently needed to support effective mitigation strategies. The uniform 1% emission factor for cropland fertilizers from the Intergovernmental Panel on Climate Change (IPCC) Tier 1 assumes that 1% of applied nitrogen-containing fertilizer is lost as N_2O. However, this emission factor cannot accurately account for local and temporal variations, owing to prevalent N_2O emission hotspots and hot moments. Process-based models also struggle with these pulsed emissions, owing to insufficient representation of N_2O production pathways, particularly the lack of separate controls over nitrifier nitrification (NN) and nitrifier denitrification (ND)[13]. Furthermore, microbial functional genes might also regulate N_2O emissions in emerging sources such as microplastic biofilms[14]. Few process-based models have integrated microbial interactions, leading to potential misattribution of sources and thus mitigation strategies being compromised.

In this Review, we discuss the sources and mechanisms of N_2O production and outline possible mitigation strategies. First, we examine hotspots and hot moments of N_2O emissions across the pedosphere, hydrosphere, atmosphere, cryosphere, lithosphere, plastisphere and industry. Subsequently, we investigate the abiotic and biotic processes that produce N_2O, and the possibility of using bioaugmentation and biogeoengineering technologies to mitigate emissions from agricultural soils and terrestrial water bodies. Finally, we identify areas for future work to address uncertainties in N_2O budgets, including the development of a global N_2O database and the application of artificial intelligence (AI) technology for N_2O source control.

Global N_2O sources

The pedosphere is the largest N_2O emission source, with agricultural and forestry ecosystems being prominent hotspots, followed by the hydrosphere and industry (Fig. 2). Spatiotemporal patterns introduce substantial uncertainties in global N_2O budget estimates. For example, cryosphere emissions are generally low but can spike during freeze–thaw cycles, and dry–wet cycles in natural grassland and desert systems can trigger episodic N_2O emission pulses. Additionally, N_2O production from the lithosphere and plastisphere might have been overlooked in global N_2O budgets. This section outlines the contributions of these ecosystems to global N_2O emissions and sources of uncertainties in existing estimates.

Pedosphere

Soils contribute more than half of terrestrial N_2O emissions (6.4 Tg N yr^{-1}, with a range of 3.8–8.7 Tg N yr^{-1})[5,6]. Hotspot emissions originate from croplands, tropical forests and pastures, whereas hot moment emissions occur in post-deforestation areas, grasslands, croplands and deserts (Fig. 2). The largest uncertainties occur in quantifying soil N_2O emissions from tropical forests and agricultural soils, owing to the complex responses of microbial soil nitrogen cycling to climate change[15], elevated atmospheric CO_2 (ref. 16), human disturbances[17] and wildfires[18].

Cropland. Agricultural soils are major N_2O emission hotspots (2.7–4.8 Tg N yr^{-1}), and N_2O hot moments occur in upland fields after fertilization (week scale)[19], irrigation and rainfall (day scale)[20], and seasonal freeze–thaw cycles in temperate regions (month scale)[21], in which spring thaw mobilizes trapped N_2O and enhances microbial activity (Table 1 and Fig. 2). Factors such as regional climate, management practices (such as nitrogen inputs, tillage, grazing and legume use), and physical and biochemical soil conditions[22–24] influence the emission of N_2O from agricultural soils, leading to high spatial and temporal variability[21]. Hot moments of episodic N_2O hotspots — locations that release large pulses of N_2O — are often associated with changes in soil moisture that stimulate the activities of nitrifiers and denitrifiers[25,26]. In contrast, N_2O cold spots — areas that consistently release low levels of N_2O — are often characterized by low dissolved organic carbon and nitrate (NO_3^-) concentrations[27].

The effect of these environmental factors on emission factors or N_2O production processes is inadequately represented, leading to considerable uncertainties in estimates of N_2O emissions from cropland soils[5]. These uncertainties are particularly pronounced in hotspot croplands such as those in eastern China, northern India and the United States Corn Belt, which have emission factors with high spatial and temporal heterogeneity[28,29]. N_2O emissions from specific croplands in these areas could be 50–100% higher than estimates using the IPCC Tier 1 default emission factor of 1% for direct emissions[23,30,31].

Forests. Forests constitute another main source (5.64 Tg N yr^{-1}) of terrestrial N_2O emissions[5]. The highest hotspots and uncertainties are for N_2O emissions from tropical regions, including the Congo Basin, the Amazon Basin and Southeast Asia[5,32]. Forest management practices, which occur across almost half of global forested areas, complicate estimates of soil N_2O emissions. Practices such as thinning, clearcutting

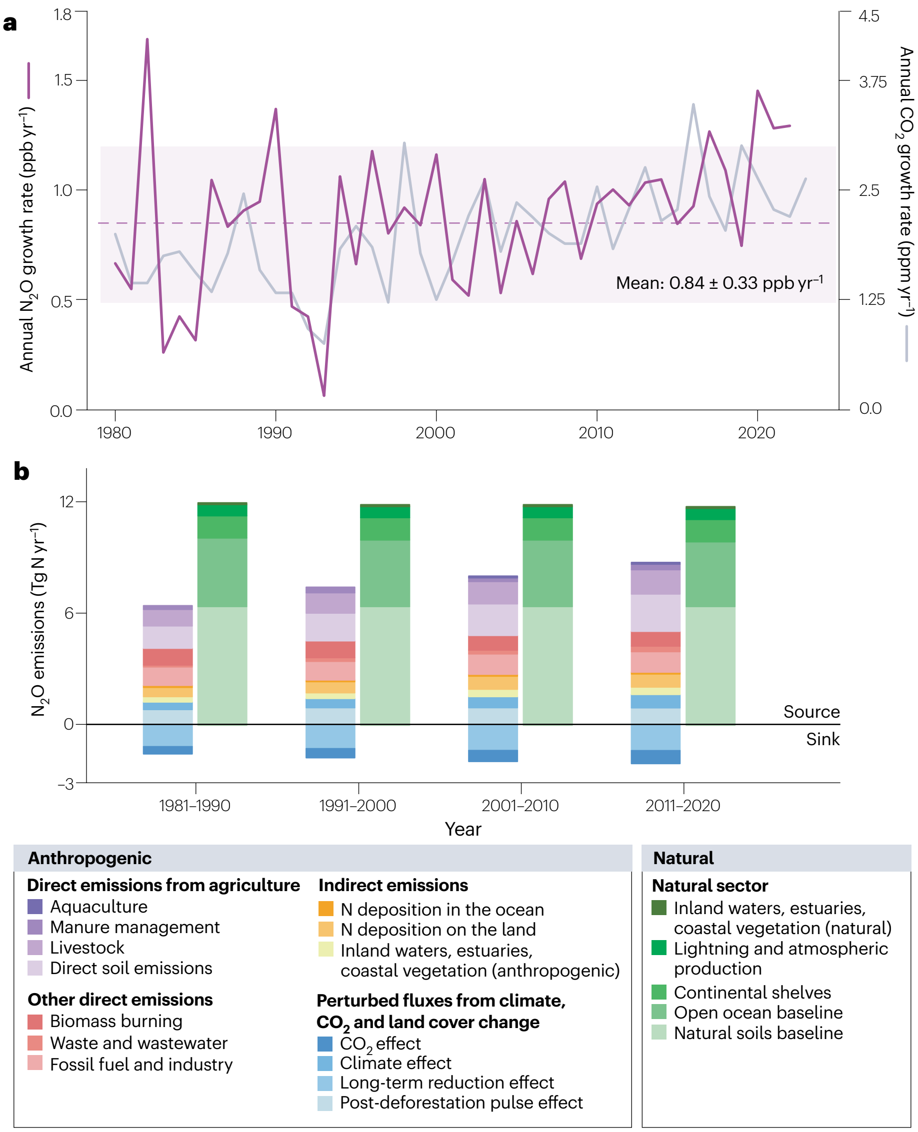
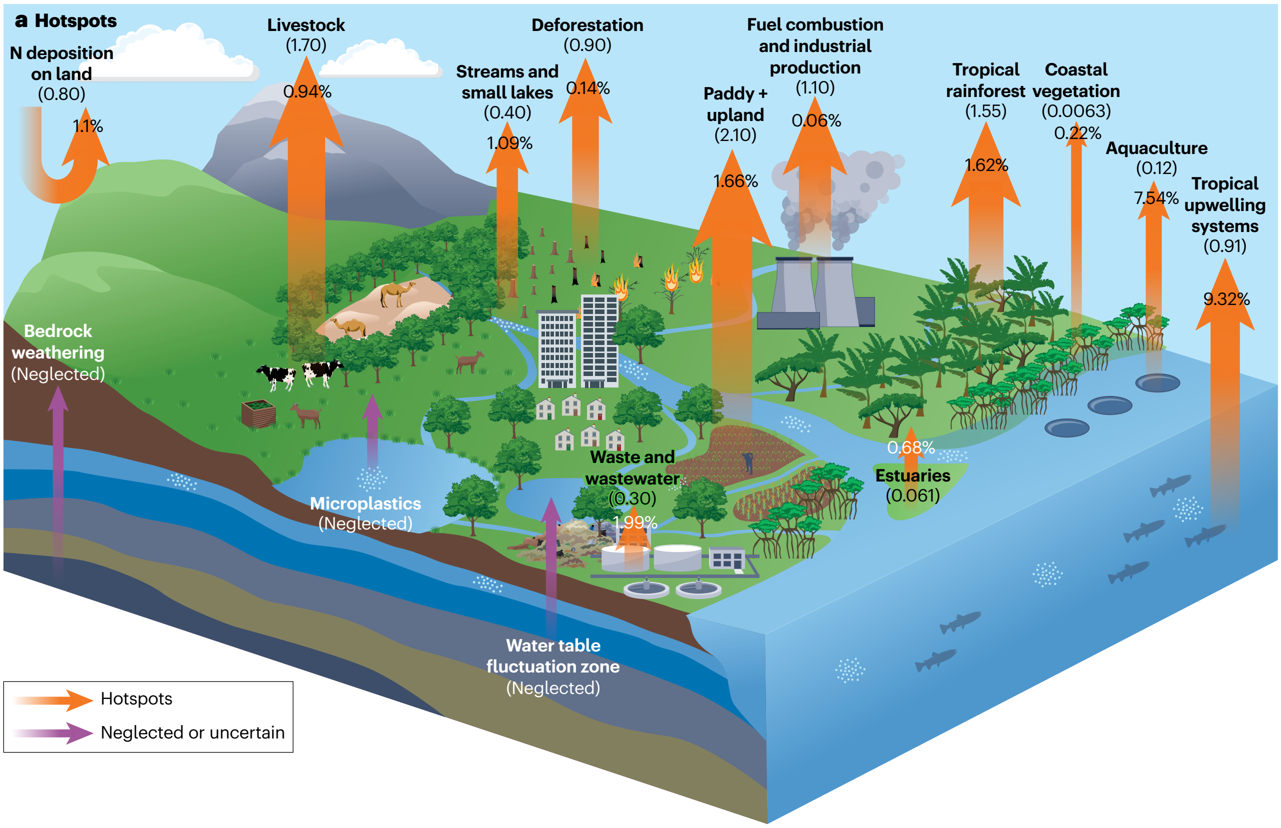


Fig. 1 | Global N_2O emission trends. a, Annual growth rates for CO_2 (grey line, 1980–2024, data from ref. 235) and N_2O (purple line, 1980–2022, data from ref. 236) based on atmospheric monitoring data. The dashed line and shaded region indicate the mean and standard deviation of the N_2O growth rate. **b**, The average annual contributions of various anthropogenic (left column) and natural (right column) factors to global N_2O emissions across four decades[5]. Global atmospheric N_2O concentrations and emissions have continued rising from 1980 to 2024, primarily driven by anthropogenic sources.

and deforestation can lead to a temporary surge in N_2O emissions from forest soil, lasting from a few months to several years, or even a decade or more[33–35] (Table 1). The post-deforestation N_2O emissions induced by this pulse effect during 2010–2019 were estimated to be around 0.9 Tg N yr^{-1} (ref. 5). However, the underlying mechanisms of these emissions remain poorly understood.

Desert and grassland. Ecosystems with dry soils, such as deserts and grasslands, are not considered major N_2O sources because they lack the wet conditions, which results in low microbial activity. However, precipitation events in grasslands following droughts can trigger hot moments of N_2O emissions (day scale) with over 200 times higher N_2O emissions than pre-drought levels[36–38]. Similarly, precipitation in dry desert soils can cause high instantaneous N_2O emission rates which can quickly increase from 0 to 1,000 ng N m^{-2} s^{-1} (hour scale)[12,39]. The projected increase in the frequency and intensity of rainfall events in deserts owing to climate change could amplify the magnitude of N_2O emission hot moments. Moreover, in northern grasslands, seasonal and diurnal freeze–thaw cycles further amplify month-scale N_2O pulses during spring thaw. The hot moments are driven by ice lens formation, substrate release and transient anaerobic microsites[11], and can contribute up to 72% of the annual emissions from northern grasslands[40]. The rising incidence of extreme climate events such as extreme rainfall, heat waves and droughts threatens the stability of soil ecosystems, resulting in substantial and unpredictable changes in N_2O emissions[41].

Global N_2O emissions from grassland are estimated to be approximately 2.2 Tg N yr^{-1}. Pasture, which refers to natural or artificial grasslands with high grazing intensity and strong human management activities such as fertilization and watering[42], has contributed up to 86% of total grassland emissions since 1961 (ref. 43). This hotspot is primarily driven by the return of animal waste to soils[36], and the strong heterogeneous emission levels are correlated with grazing intensity and pasture management practices[44] (Table 1). The mobile excretion behaviour of grazing animals exacerbates the spatial variability of


a Hotspots
N deposition on land (0.80)
1.1%
Livestock (1.70)
0.94%
Deforestation (0.90)
0.14%
Streams and small lakes (0.40)
1.09%
Fuel combustion and industrial production (1.10)
0.06%
Paddy + upland (2.10)
1.66%
Tropical rainforest (1.55)
1.62%
Coastal vegetation (0.0063)
0.22%
Aquaculture (0.12)
7.54%
Tropical upwelling systems (0.91)
9.32%
Bedrock weathering (Neglected)
Microplastics (Neglected)
Waste and wastewater (0.30)
1.99%
0.68%
Estuaries (0.061)
Water table fluctuation zone (Neglected)
Hotspots
Neglected or uncertain

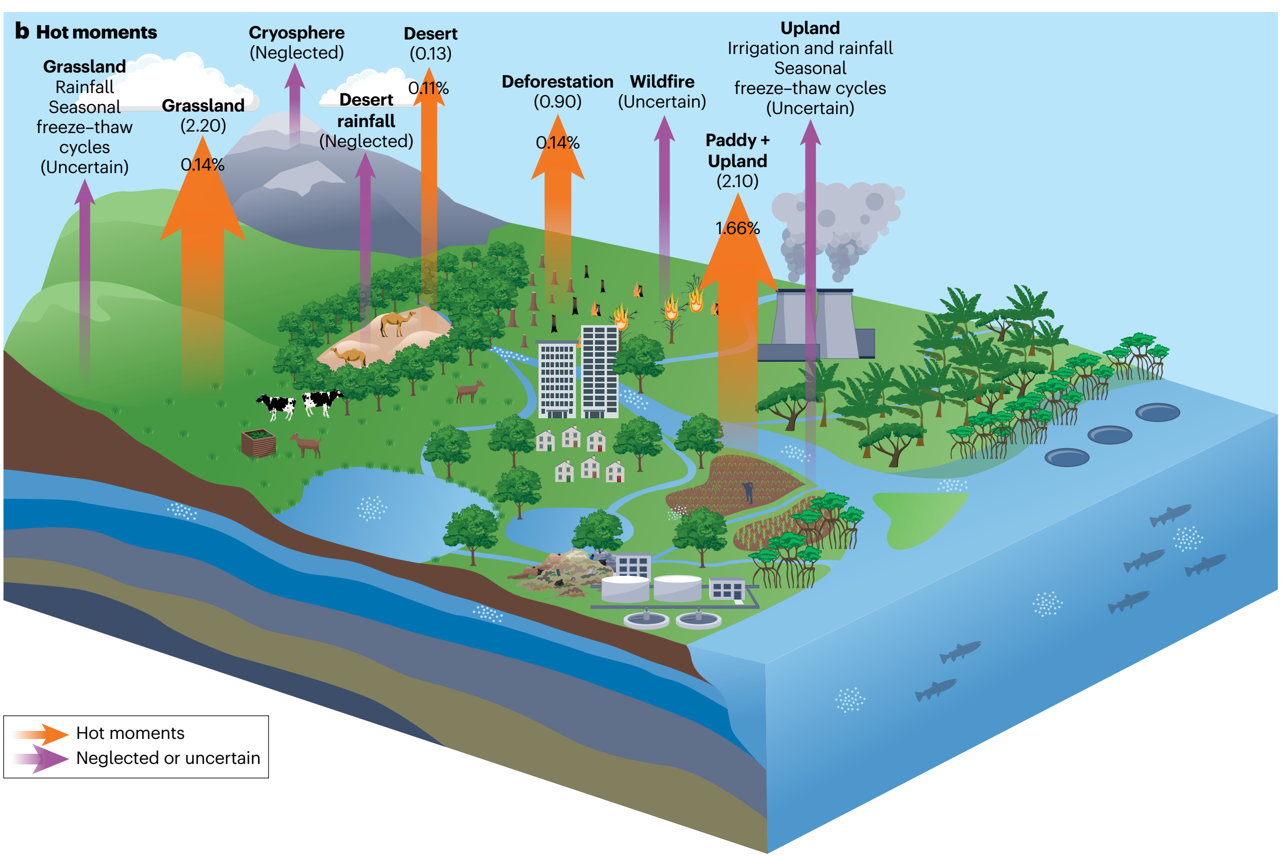

b Hot moments
Grassland
Rainfall
Seasonal freeze–thaw cycles
(Uncertain)
Grassland (2.20)
0.14%
Cryosphere (Neglected)
Desert (0.13)
0.11%
Desert rainfall (Neglected)
Deforestation (0.90)
0.14%
Wildfire (Uncertain)
Upland
Irrigation and rainfall
Seasonal freeze–thaw cycles
(Uncertain)
Paddy + Upland (2.10)
1.66%
Hot moments
Neglected or uncertain

Fig. 2 | Global N_2O emission hotspots and hot moments. a, The global fluxes (shown in brackets, in Tg N yr^{-1}) and average annual growth rates of N_2O emissions (shown on the arrow, %) across various anthropogenic and natural emission hotspots (orange arrows): nitrogen deposition on land, livestock, streams and small lakes, paddy fields and upland, fuel combustion and industrial production, and waste and wastewater (flux calculated for 2020 (ref. 5); growth rate calculated for 1980–2020 from Table 1); deforestation (flux: 2010–2019 (ref. 5); growth rate: 1980–2020 from Table 1); tropical rainforest[237,238] (flux: 1991–2000; growth rate 1995–2012); coastal vegetation[239,240] (flux: model estimate; growth rate: 2014–2024); aquaculture (flux: 2020 (refs. 241,242); growth rate: 1980–2020 from Table 1); tropical upwelling systems (flux: model estimate[49]; growth rate: 2003–2020 (ref. 243)); and estuaries[59,244] (flux: model estimate; growth rate: 1998–2022). Average annual growth rates taken from Table 1 are the average of the growth rate in each year in the period. Those for tropical rainforest, coastal vegetation, estuaries and tropical upwelling systems are calculated through the following formula: growth rate = $(\frac{\text{final flux}}{\text{initial flux}})^{\frac{1}{\text{years}}} - 1$. Arrow thickness reflects the N_2O emission magnitude. Purple arrows indicate sources that are neglected or uncertain in global climate models and the N_2O budget: bedrock weathering[87], microplastics, and water table fluctuation zone. **b**, As in **a** but for hot moments in grassland (flux: 2006 (ref. 44); growth rate: 1980–2020 (ref. 245)), desert[246,247] (flux (shown in kg N ha^{-1} yr^{-1}) and growth rate: 2014–2017), deforestation (flux: 2010–2019 (ref. 5); growth rate: 1980–2020 from Table 1), and paddy and upland (flux: 2020 (ref. 5); growth rate: 1980–2020 from Table 1). Annual growth rate values for desert and grassland are directly obtained from the references. The contributions are uncertain or neglected for: grassland from rainfall and seasonal freeze–thaw cycles; the cryosphere; desert rainfall; wildfire; and upland from fertilization, irrigation, rainfall and seasonal freeze–thaw cycles. Individual ecosystems make various contributions to global N_2O emissions; however, many sources remain neglected or uncertain in existing research estimates.

field-scale N_2O emissions[45]. For example, in hill-grazing systems, livestock predominantly congregate in topographically low-lying areas, leading to the localized accumulation of urine and manure deposits, which amplifies the N_2O emission hotspot[45,46]. Emissions can be mitigated by moving livestock across the landscape to ensure an even distribution of manure nitrogen, enhancing soil nitrogen retention and increasing the efficiency of nitrogen uptake by plants[47,48].

Hydrosphere

N_2O emissions from continental shelves and the open ocean remained relatively stable between 1980 and 2020 (on average 4.2 Tg N yr^{-1} according to ref. 49 or on average 4.8 Tg N yr^{-1} according to ref. 5). Tropical waters are assumed to be N_2O emission hotspots, contributing about 64% of global ocean N_2O emissions. Coastal upwelling systems contribute another 20% (ref. 49) despite occupying less than 3% of the ocean area. There are large uncertainties in oceanic N_2O budgets because it is unclear how the expansion of oxygen minimum zones[50] and the ongoing global ocean deoxygenation[51] caused by rising nutrient loads and climate change will affect N_2O emissions. Additionally, seasonal stratification is likely to lead to heterogeneity in the N_2O emissions of natural aquatic ecosystems. In summer, water stratification creates distinct oxycline and thermocline, with anoxic conditions in hypolimnion promoting N_2O production[52]. As water mixes in autumn and winter, the hypolimnetic N_2O is transferred to the epilimnion, increasing N_2O emissions and creating a seasonal flux pattern. Sunlight intensity and duration also influence the N_2O yielded from photochemodenitrification[53], leading to diurnal N_2O emission patterns.

Human activities are driving an increase in N_2O emissions from rivers, lakes, reservoirs, estuaries and coastal vegetation, mainly through nitrogen loss to inland waters through agricultural activities. In 2010, the N_2O emissions from all types of inland waters (such as streams, rivers, lakes and reservoirs) were estimated to be 0.32 ± 0.01 Tg N yr^{-1}, with streams and small lakes being the main hotspots[54–56]. It was estimated that these emissions had increased to 0.51 ± 0.22 by 2020. Global riverine N_2O emissions have increased fourfold since 1900, mainly owing to agricultural nitrogen applications[55] (Table 1). Additionally, N_2O emissions from global lentic systems (lakes and reservoirs) have increased by 126% since the 1850s, owing to climate change and human disturbances[54]. The largest uncertainty of N_2O emissions arises from stream and river networks[54]. The heterogeneity of N_2O emissions from inland waters is primarily manifested in riparian zone systems[8,57,58], which are transitional boundary zones between terrestrial and aquatic ecosystems and have an important role in regulating landscape-level interactions. Estimates of global estuarine N_2O emissions are highly uncertain (0.04–3.63 Tg N yr^{-1} (refs. 59–62)) (Table 1), owing to the variable environmental conditions, such as salinity and changeable nutrient inputs. Tidal systems and deltas (35 Gg N_2O yr^{-1}) make the largest contribution to estuarine N_2O emissions (61 Gg N_2O yr^{-1})[60]. Mangroves are the largest N_2O source (8.9 Gg N_2O yr^{-1}) for coastal ecosystems, whereas seagrasses are a major sink (4.6 Gg N_2O yr^{-1}).

N_2O emissions from aquaculture (average annual growth rate 7.54%) and wastewater (1.99%) are increasing by 0.1 Tg N per decade, making them the fastest-growing sources in the hydrosphere[60]. Traditional wastewater treatment using nitrification and denitrification produces ~3–7% of global N_2O emissions, with a consistent annual increase of 0.04 Tg N yr^{-1} from 1980 to 2015 (refs. 6,63). This trend is projected to continue, owing to the expansion of wastewater infrastructure driven by population growth and urbanization[64]. Traditional wastewater treatment processes also consume a large amount of energy for aeration and pump circulation.

Indirect emissions from agricultural N_2O transcend ecosystem and sphere boundaries, creating hotspots and hot moments. Hydrological connections can rapidly transport N_2O stored in soils and groundwater into streams where it is then released into the atmosphere[55,65–67], creating hot moments, particularly in areas with enhanced drainage, such as the tile-drained agricultural fields of the US Corn Belt and similar systems in Europe[68]. Similarly, globally prevalent karst landscapes, which rapidly transport water, create N_2O emission hotspots[67,69]. N_2O concentrations are often an order of magnitude higher in groundwater than in surface water; therefore, these areas are likely to constitute a major, yet poorly characterized, source of uncertainty for global N_2O estimates.

Atmosphere

Although the atmosphere is a N_2O sink, it is an important part of the global budget. The primary removal mechanisms of atmospheric N_2O are photolysis in the stratosphere[70] and reactions with hydroxyl radicals in the troposphere[71]. N_2O has a long atmospheric lifespan (116 ± 9 years (ref. 72)). Additionally, anthropogenic inputs of reactive nitrogen into the global nitrogen cycle have caused the rate of increase of atmospheric N_2O to rise from 0.22 ± 0.07 ppb yr^{-1} during 1901–1964 to 0.69 ± 0.33 ppb yr^{-1} during 1966–2008, with an even faster acceleration since 2009 (0.89 ± 0.21 ppb yr^{-1} during 2009–2021 (ref. 4)).

Table 1 | Anthropogenic N_2O emissions

Source	Flux[5] ($Tg\,N\,yr^{-1}$)[a]	Average annual growth rate[5] (%)[b]	Driving factors[c]	Source of uncertainties	Pathways	Hot moment or hotspot	Projected impact of climate change
Direct emissions from agriculture							
Paddy	1.2–2.1[d]	1.66	Fertilization, dry–wet cycle; pH ($NH_4^+ < NO_3^-$)	Water management; organic fertilizer type	Nitrification and denitrification	Tillage layer	Shorten rice growth cycle; increase nitrogen turnover rate
Upland			Fertilization; moisture; cultivation ($NH_4^+ > NO_3^-$)	Fertilization and irrigation; manure and crop residues	Microbial process; NH_4^+ driven	After fertilization or irrigation	Increase ammonia volatilization; increase wet deposition and nitrogen loading
Livestock	1.1–1.7[e]	0.94	Grazing ($NH_4^+ > NO_3^-$)	Breeding technology and management	Nitrification and denitrification	Pasture soils with animal waste return	Increase activity of nitrifiers and denitrifiers; increase availability of soil carbon and nitrogen
Aquaculture	0.01–0.12[f]	7.54	Feed supply (NH_4^+ and NO_3^-)	Breeding technology and scale; management	Microbial process; denitrification	Surface sediment	Accelerate feed consumption and nitrogen input
Direct emissions from industry							
Fuel combustion and industrial production	1.0–1.1[g]	0.06	Legal regulations and supervision (NH_3)	Implementation and enforceability	Catalytic oxidation of ammonia	Unknown	Not directly related
Waste and wastewater	0.1–0.3[h]	1.99	Diffused and point source pollution (NH_4^+ and NO_3^-)	Treatment technology	Nitrification and denitrification	Oxic processes	Increase microbial processes and ammonia volatilization; degrade wastes into simple carbon sources
Indirect emissions							
Nitrogen deposition on land	0.5–0.8[e]	1.10	Agricultural and industrial intensity ($NH_4^+ > NO_3^-$)	Climate change; human disturbances	Nitrification and denitrification	China, South Asia and Southeast Asia	Increase nitrogen evaporation and precipitation; stimulate microbial activity and abundance
Inland water, estuaries, coastal vegetation	0.3–0.4[h]	1.09	Diffused and internal pollution; dissolved oxygen ($NH_4^+ > NO_3^-$)	Spatial heterogeneity; estimation methods	Ammonia oxidation; nitrifier denitrification	Streams and small lakes	Increase microbial processes and evaporation rates; promote nitrogen conservation
Perturbation fluxes							
Post-deforestation	0.9[e]	0.14	Rising temperatures, soil microbiology (unknown)	Fire; estimation methods; lack of data	Nitrification and denitrification	After large-scale forest logging	Reduce canopy interception; accelerate molecular motion; enhance microbial activity
Microplastics	Unknown		Plastic abundance (NH_4^+ and NO_3^-)	Unknown	Microbial processes	Unknown	Accelerate microplastic decomposition; increase microbial activity

[a]Values for 1980–2020, except post-deforestation flux, which is for 2010–2019. [b]The average annual growth rate (1980–2020) is obtained by calculating the annual growth rate of N_2O emission flux each year and then averaging the results. [c]Driving nitrogen compounds in brackets. [d]Measured with closed chamber technique and eddy covariance. [e]Measured with closed chamber technique. [f]Measured with eddy covariance and floating chamber method. [g]Measured with online nitrogen oxide monitoring system. [h]Measured with floating chamber method.

Consequently, atmospheric N_2O concentrations have increased to 333 ppb in 2021 from a preindustrial value of ~270 ppb (ref. 73).

Since the early twenty-first century, N_2O has become a dominant ozone-depleting substance[2], further complicating climate system feedback. Comparing observations of atmospheric N_2O concentrations with the results of terrestrial biosphere models suggests that land N_2O emissions strongly influence the seasonal amplitude of tropospheric N_2O concentrations[74], and this amplitude is modulated by the intrusion of N_2O-depleted air from the stratosphere[75]. Concurrently, climate change is causing more vigorous stratospheric circulation, leading to a reduction in the atmospheric lifetime of N_2O (−2.1 ± 1.2% per decade), which could mitigate the impact of N_2O on anthropogenic warming and ozone depletion[76]. This interaction creates a bidirectional feedback loop: N_2O emissions drive radiative forcing, while climate-driven atmospheric dynamics modulate the persistence and global distribution of atmospheric N_2O. Further exploration is required to assess how the sources and sinks of N_2O in the atmosphere will be affected by climate change and to explore the associated feedbacks to the climate system.

Cryosphere

N_2O emissions from permafrost-affected peatland soils in the Northern Hemisphere have not yet been considered in global emission budgets and models, but they are becoming substantial sources of N_2O. These sources have historically been considered negligible because of assumptions of nitrogen limitation[77–80]. However, thawing permafrost in the Northern Hemisphere has been found to act as a previously underestimated N_2O source, with estimates ranging from 30 to 455 μg N m^{-2} d^{-1} (refs. 21,77,80,81). Thus, it is important to consider permafrost-affected soils in Earth's N_2O budget, particularly because these regions are warming at a rate of 0.6 °C per decade, which is twice the global average. Ice-covered and snow-covered landscapes (such as permafrost peatlands) could also act as emission hotspots during episodic thaw events, with microbial activity in anaerobic microsites driving pulsed N_2O release[82]. Additionally, seasonally frozen peatlands exhibit a 2-month-scale hot moment of N_2O emissions during the thawing period[11].

Although seasonal freeze–thaw cycles in non-permafrost systems (such as croplands and grasslands) generate month-scale N_2O hot moments[40,83,84], these are partially accounted for in regional inventories[21,40,83,84]. By contrast, emissions from permafrost degradation — where soils remain frozen for most of the year — are not yet represented in global emission budgets or models, despite their growing climate feedback potential[21,82].

Lithosphere

Bedrock represents the largest terrestrial nitrogen reservoir[85,86]; however, bedrock weathering has not been incorporated in process-based models. There is currently limited research on biological N_2O production in bedrock; instead, research focuses on biological N_2O production from the release of nitrogen in downstream soil or water environments. Physical and chemical weathering processes, which are accelerated by microorganisms, especially fungi[86], release nitrogen to the terrestrial surface and subsurface environments. The released nitrogen is nitrified to NO_3^-, which is then denitrified in surface water and groundwater, with N_2O as an intermediate product during the process[87]. Global estimates indicate that about 10.0–18.0 Tg weathered-nitrogen is released each year[86], with bedrock weathering accounting for approximately 10–20% of the terrestrial nitrogen flux associated with N_2O (ref. 87). The seasonally fluctuating water table results in an emission hotspot associated with the release of bedrock nitrogen[87].

Plastisphere

The microbial community residing on plastic debris, known as the plastisphere, spans multiple biomes and can drive N_2O production[88]. The microbial biomass associated with the plastisphere is substantial[89]. The formation of biofilms on the surfaces of plastic debris creates oxic–anoxic microenvironments[14,90,91] that influence element cycling and N_2O production[92]. The presence of microplastics in oceans, freshwater bodies and soils increases N_2O production by 10–70%. This increase is primarily caused by nitrification and denitrification processes, but it is unclear what factors lead to N_2O hot moments from plastics[93–96] (Table 1). It is expected that N_2O production from the plastisphere will continue to rise with the ongoing increase in global plastic pollution[83]. This contribution to N_2O emissions has rarely been considered in relevant global assessments.

Industry

Industry is responsible for about 14% of global anthropogenic N_2O emissions. These emissions are primarily unintended by-products of various industrial processes, including fossil fuel combustion, the production of adipic acid and nitric acid, and smaller contributions from the production of caprolactam, glyoxal and niacin, which are not affected by climate change[97] (Table 1). Industrial N_2O production primarily involves chemical oxidation and reduction reactions at high temperatures; for example $N_2 + O_2 \rightarrow 2NO$ and $2NO + N_2 \rightarrow 2N_2O$. Technologies such as thermal destruction, catalytic destruction and catalytic reduction can reduce these N_2O emissions. The use of these technologies by many operating plants to implement mitigation measures during 1990–1995 led to a substantial reduction in N_2O emissions. However, emissions increased sharply after 2010 as the implementation of emission-reducing strategies declined[6]. The reduction of industrial N_2O emissions mainly relies on governance and management, aided by the availability of low-cost techniques.

Uncertainties

Scale effects. Microscale heterogeneity affects hotspots and hot moments of N_2O emissions within diverse landscapes. Abrupt and large N_2O pulses can be triggered by soil properties, microclimate, microtopography and substrate availability[25,98,99]. Soil texture determines its thermal and hydrological conductivities, thereby regulating the capacity for nitrification and denitrification. For example, sandy agricultural soils, characterized by high porosity and low water retention capacity, promote rapid soil temperature change, drainage and oxygen diffusion, which enhance nitrification but limit denitrification, leading to pronounced spatial heterogeneity of N_2O emissions[99]. Soil oxygen availability also acts as a primary regulator, with denitrification dominating in low-O_2 zones (such as waterlogged soils, clay-rich environments or the rhizosphere, where O_2 is depleted by root and microbial respiration), and nitrification prevailing in aerobic microsites (for example well-drained sandy soils)[100,101]. Microtopographic features can create emission hotspots by altering microclimates[68]. For instance, increased solar radiation, owing to forest canopy gaps, elevates soil temperature, which boosts microbial activities, leading to enhanced N_2O production. Conversely, reduced litter inputs disrupt carbon–nitrogen coupling and potentially increase N_2O–N_2 ratios[102]. Hydrological connectivity in small streams and hyporheic zones further amplifies emissions through wet–dry cycles that stimulate coupled nitrification–denitrification[56].

Existing mechanistic models do not soundly incorporate these processes and therefore cannot effectively capture emission pulses. Integrating scale-adaptive processes and developing fine-resolution environmental input data could improve the ability of these models to capture episodic N_2O emissions at microscales[55].

Anthropogenic emissions. The uncertainty surrounding N_2O emissions from most ecosystems is mainly driven by the sporadic nature of human disturbances. Other sources of uncertainty include variations in the technical methods used for measurements in different regions; excessive fluctuations in flux emissions owing to human disturbance or frequent management; and the lack of data for many regions or new emission sources (Table 1). To reduce this uncertainty, it is necessary to understand the processes that produce N_2O and the underlying mechanisms; how process dynamics influence uncertainty and the ability of existing tools to capture hotspots, hot moments and dynamic fluxes; and whether it is possible to measure N_2O fluxes between pools and what limitations hinder the development of better models.

Biogeochemical N_2O production processes

To reduce uncertainty in N_2O emissions across different ecosystems, it is important to elucidate the dominant N_2O production processes and characterize their dynamic responses to environmental changes. In this section, we outline the biogeochemical mechanisms of N_2O production and the uncertainties in these mechanisms. This knowledge is essential for developing technologies and identifying key control parameters to manage and reduce N_2O emissions.

Biotic N_2O production

Biotic N_2O production involves intricate biogeochemical processes that are largely driven by microorganisms. There are two distinct microbial processes (Fig. 3). First, there is ambient ammonium (NH_4^+)-derived N_2O production (NH_3 as substrate), which occurs through three mechanisms: nitrifier nitrification (NN), nitrifier denitrification (ND) and nitrification-coupled denitrification (NCD)[103]. Second, there is ambient NO_3^--derived N_2O production (NO_3^- as substrate), which is attributed to denitrifier denitrification (DD). The NN process primarily relies on the enzymes ammonia monooxygenase (AMO, *amoA/B/C*) and hydroxylamine dehydrogenase (HAO, *haoA/B*) for ammonia (NH_3) oxidation. Conversely, the ND process mainly involves copper-containing nitrite (NO_2^-) reductase (Cu-NIR, *nirK*) and cytochrome *c*-dependent nitric oxide reductase (cNOR, *cnorB/C*) for NO_2^- reduction and NO reduction, respectively. The NCD and DD processes share key enzymes, including cd_1-containing NO_2^- reductase (cd_1-NIR, *nirS*), Cu-NIR, cNOR and N_2O reductase (NOS, *nosZ*). The fundamental difference between these processes is that in the NCD mechanism, the NO_3^- substrate is sourced from NH_3 oxidation, whereas in DD, NO_3^- is derived directly from the environment. There is another distinct enzymatic pathway in which the periplasmic cytochrome (cyt) P460 enzyme facilitates the oxidation of two molecules of hydroxylamine (NH_2OH) to N_2O under anoxic conditions[104]. Furthermore, fungi carrying cytochrome P450nor NO reductase can yield N_2O as a terminal product[105].

NO_3^--derived N_2O production. Denitrification has a pivotal role in the global nitrogen cycle and drives NO_3^--derived N_2O production. In organic-rich environments, heterotrophic denitrifiers sequentially reduce NO_3^- to NO_2^-, NO, N_2O and ultimately N_2 (Fig. 3). However, oligotrophic systems host denitrifiers that use inorganic electron donors such as elemental sulfur, sulfide, thiosulfate or H_2 to reduce NO_3^- to N_2 (refs. 106,107). Methane-dependent denitrifiers further expand this metabolic diversity, using CH_4 as an electron donor for NO_3^- and NO_2^- reduction[108]. Bacterial denitrification proceeds in a modular manner, and the final step, reduction of N_2O to N_2, is often incomplete, leading to N_2O accumulation[109].

NO_3^--derived N_2O production is regulated by environmental factors such as oxygen availability, pH, temperature and trace metals (for example copper ions). Physiological investigations indicate that high O_2 concentrations decrease the rate of microbial N_2O reduction[110]; however, microcosm experiments suggest that prolonged O_2 exposure induces a denitrification phenotype with accelerated N_2O reduction[111]. It will be important to better understand the effect of O_2 on denitrification to predict N_2O emissions from habitats with fluctuating O_2 concentrations[111]. Soil moisture dynamics also exert strong control: alternating wetting–drying cycles in permafrost zones[112], peatlands[113] and global paddy systems (for example in the Indian subcontinent[114], North China Plain[115], Bangladesh[116] and Finland[117]) lead to a 30–45 times increase in N_2O emissions compared with continuously flooded soils. Although warming directly stimulates denitrification, it is unclear whether it amplifies net N_2O production or consumption[118,119]. Historically, acidic pH was thought to prevent the microbial conversion of N_2O to N_2; however, the discovery of acid-tolerant N_2O-reducing microorganisms capable of efficient N_2O reduction under low pH conditions challenges this paradigm[120,121].

NH_4^+-derived N_2O production. NH_4^+-derived N_2O production is promoted by factors such as good substrate availability, NO_2^- detoxification, and the absence of N_2O reduction activity. The enzymes involved in the NN and ND pathways − AMO and HAO − are localized in the cytoplasmic membrane and periplasm of the nitrifiers[122,123]. This spatial organization enables the ND pathway to directly use NO_2^- generated by NN, creating a synergistic relationship that minimizes energy expenditure and avoids competition with other microorganisms[124]. Furthermore, NO_2^- reduction via ND alleviates NO_2^- toxicity (a biological detoxification mechanism[125]), increasing the activity of the NN pathway. Nitrifier genomes lack N_2O reductase gene homologues, preventing the enzymatic reduction of N_2O to N_2. Consequently, N_2O generated by NN and ND is released rather than metabolized within the same cell[126,127].

NH_4^+-derived N_2O production pathways are mediated by three microbial groups: ammonia-oxidizing bacteria (AOB), ammonia-oxidizing archaea (AOA), and complete ammonia oxidizers (comammox) (Fig. 3b, left). AOB host both a NOR enzyme (existing in two genetic variants encoded by *nor*CBQD or *nor*SY[128,129]) and cyt P460, enabling the production of N_2O via the ND and cyt P460 mediated pathways. In contrast, AOA lack canonical *nor* genes but possess a putative cytochrome P450nor NO reductase gene, which might facilitate ND-driven N_2O production in acidic environments[130]. Additionally, N_2O production catalysed by a multicopper enzyme unique to AOA has been demonstrated[131], elucidating an AOA mediated biotic process for generating N_2O. Consequently, AOA produce N_2O in a hybrid mode (biotic and abiotic pathways)[132].

Comammox bacteria challenge the traditional understanding that NH_4^+ oxidation to NO_2^- and NO_3^- requires sequential steps mediated by distinct microbial groups[122]. Molecular biology analysis confirms the wide distribution of the comammox bacteria in diverse environments, including soils, rivers, lakes, tidal flats, estuaries, and engineered systems such as wastewater treatment plants, drinking water systems and aquaculture facilities[133]. Comammox bacteria have a high affinity for NH_3, low growth rate and high growth yield compared with canonical NH_3 oxidizers (AOA and AOB)[134]. Their genomes lack *nor* homologs, and although cytochrome *c*′-β (encoded by *cytS*) has been detected sporadically in *Nitrospira* species, its functional role in N_2O production remains unconfirmed[135]. However, comammox bacteria can indirectly contribute to abiotic N_2O formation by supplying NH_2OH, a reactive intermediate[135]. The use of comammox-dominant NH_3 removal processes for practical applications remains limited, but such applications might emerge as their ecological and engineering potential is further explored.

The sensitivity of NH_4^+-derived N_2O production to climate change remains unclear because nitrifiers respond in complex ways to environmental shifts. Elevated temperature alone enhances nitrifier activity,

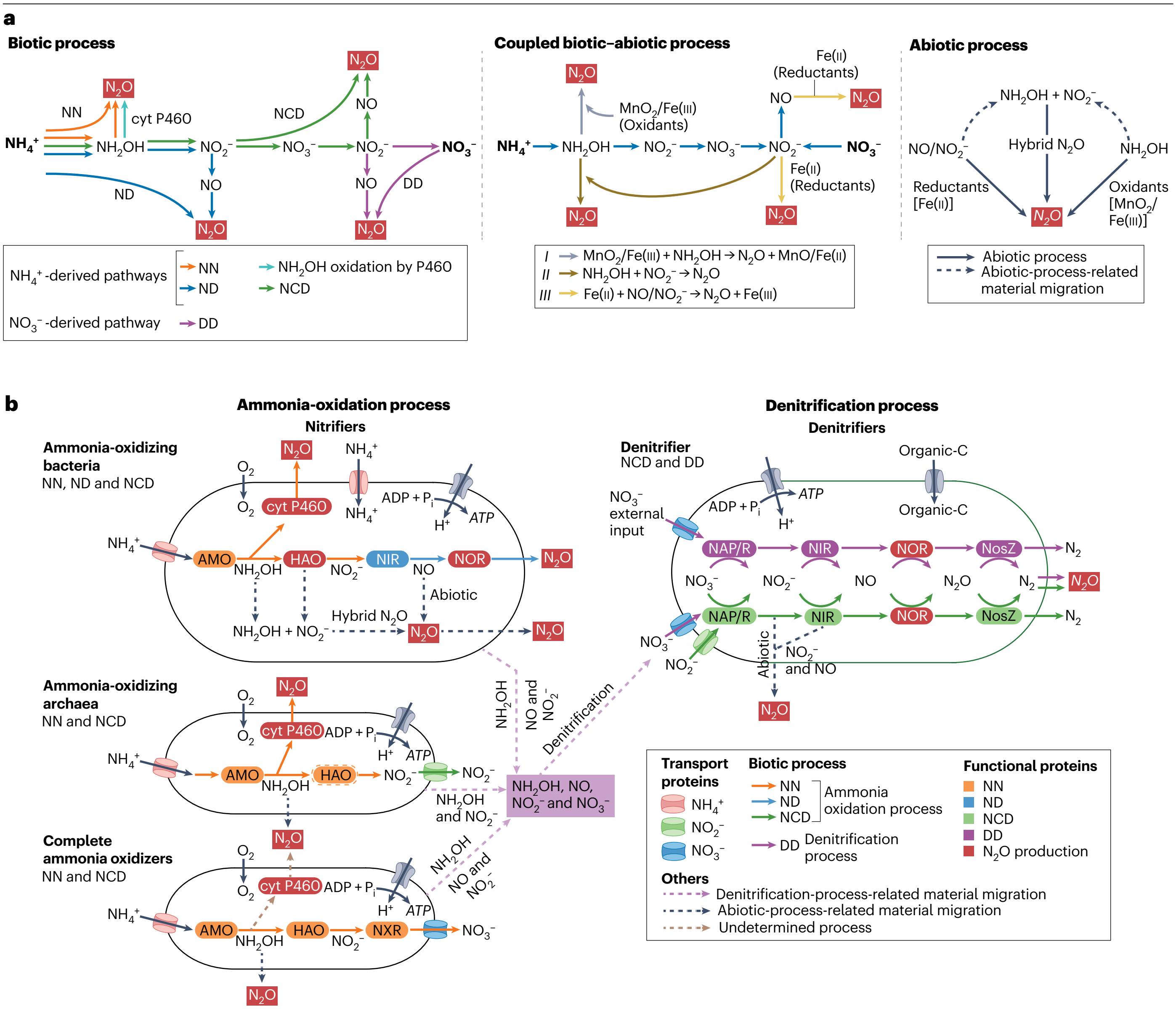


Fig. 3 | N_2O production pathways. a, Left, biotic N_2O production pathways and the associated substrates and enzymes. There are three processes driven by ambient NH_4^+ (nitrifier nitrification, NN; nitrifier denitrification, ND; nitrification-coupled denitrification, NCD), one process driven by ambient NO_3^- (denitrifier denitrification, DD) and an NH_2OH oxidation route catalysed by cyt P460. Middle, the coupled biotic–abiotic N_2O production occurs through a single pathway, differentiated by the source of the substrates, and can involve chemodenitrification (yellow arrows) or hydroxylamine conversion (grey and brown lines). Right, the abiotic N_2O production pathway occurs entirely through chemical reactions, involving chemodenitrification (reductants arrow) and hydroxylamine conversion (hybrid and oxidants arrows). **b**, Metabolic diagram of the ammonia oxidation and denitrification pathways associated with N_2O production, showing the transfer and transformation of electron donors and electron acceptors (solid arrows) and the transfer of substrates and undetermined processes (dashed arrows). In ammonia-oxidizing archaea, the Hao-related proteins are represented by dotted lines because they differ from those encoded by the well-characterized bacterial *hao* gene. ADP (adenosine diphosphate) + P_i (inorganic phosphate) → ATP (adenosine triphosphate) represents the storage of cellular energy. AMO, ammonia monooxygenase; HAO, hydroxylamine dehydrogenase; NAR/P, nitrate reductase; NIR, nitrite reductase; NOR, nitric oxide reductase; NXR, nitrite oxidoreductase. Microbially, the ammonia oxidation pathway is the chief N_2O-producing route, dominated by ammonia-oxidizing bacteria.

with AOA being more sensitive than AOB[136,137]. However, increased CO_2 concentrations generally inhibit most nitrifiers (AOB, AOA and NO_2^--oxidizing bacteria, NOB), except comammox[138,139]. To accurately assess the impacts of climate change on NH_4^+-derived N_2O production, it is important to consider the effects of specific ecosystem type and human influences (such as fertilizer use)[140].

Tracing formation pathways. The sources of nitrogen and oxygen vary substantially between the different N_2O production pathways[141]. The nitrogen atoms in N_2O generated via NN, ND and NCD originate entirely from NH_4^+, classifying these as NH_3-driven pathways, whereas denitrifier denitrification (DD) derives nitrogen from NO_3^-, defining it as a NO_3^--driven process. Oxygen atom sources further differentiate these pathways: in NN, oxygen atoms are exclusively sourced from O_2; in ND, half originate from O_2 and half from water; in NCD, one-third derive from O_2 and two-thirds from water; and in DD, all oxygen atoms originate from NO_3^-.

Isotope tracing techniques can use these different nitrogen and oxygen sources to elucidate the formation mechanisms of N_2O. Conventional ^{15}N single-isotope tracing approaches cannot resolve oxygen origin; however, dual ^{15}N–^{18}O isotopic tracing has been developed to identify the source of nitrogen (NH_3 or NO_3^-) and oxygen (NO_3^-, water or O_2). This technique can apportion contributions of each pathway to N_2O production by analysing isotopic excess in N_2O (refs. 142,143). However, overlapping oxygen atom sources between the different NH_3-driven processes (NN, ND, NCD) preclude precise quantification of individual contributions, limiting estimates to ranges of maximum and minimum values[142,144]. Additionally, the dissimilatory NO_3^- reduction to NH_4^+ process, which is often overlooked in isotopic assessments, introduces complexity by altering the isotopic compositions of NH_4^+ and NO_3^- pools, thereby skewing N_2O source calculations.

The site preference of ^{15}N ($\delta^{15}N^{sp}$) provides an alternative approach to identify N_2O sources and sinks across environment scales[145]. Measurements of $\delta^{15}N^{sp}$ reflect the intramolecular distribution of ^{15}N in the asymmetric N_2O molecule[146]. The value of $\delta^{15}N^{sp}$ is independent of the isotopic composition of the substrate and reflects biochemical production mechanisms and isotopic fractionation during N_2O reduction[147]. Different enzymatic pathways have distinct $\delta^{15}N^{sp}$ values. For example, fungal denitrification and NN lead to elevated $\delta^{15}N^{sp}$, owing to the sequential binding of nitrogen atoms to a single haem iron centre, whereas near-zero $\delta^{15}N^{sp}$ arises from the simultaneous binding of two nitrogen atoms to dual iron centres in DD and ND[148,149].

Together, the isotopocule signatures of $\delta^{15}N^{bulk}$, $\delta^{18}O$ and $\delta^{15}N^{sp}$ can disentangle each pathway. $\delta^{15}N^{sp}$ alone is insufficient for this task because the values overlap between processes, and multiple N_2O production and reduction pathways occur at the same time[150,151]. Isotope mapping based on isotope mixing principles constrains the extent of N_2O reduction and dominant N_2O production pathways using dual isotopocule signatures ($\delta^{18}O$/ $\delta^{15}N^{sp}$ or $\delta^{15}N^{sp}$/ $\delta^{15}N^{bulk}$)[150]. The Bayesian mixing model known as stable isotope fractionation and mixing evaluation (FRAME) integrates three isotopocule signatures to identify the most probable solution for N_2O reduction and the mixing proportions of each N_2O pathway via a Markov chain Monte Carlo algorithm[152,153]. At broader scales, isotopocule data can be integrated with biogeochemical models (such as DAYCENT and DNDC) to gain insight into the regional and global patterns of N_2O fluxes[154].

Coupled biotic–abiotic N_2O production

Besides biotic pathways, N_2O can also be produced by biotic–abiotic processes (Fig. 3). These processes are primarily influenced by factors such as pH, oxygen, temperature, organic carbon content and metal ions[155]. Coupled biotic–abiotic N_2O production has been widely detected in natural and artificial habitats, with varying contributions depending on different habitats[156,157]. However, most process-based models do not account for coupled biotic–abiotic processes as an N_2O source. In industry, N_2O can also be produced through abiotic processes; however, the main mechanism of these abiotic processes is already clearly understood and will not be discussed here[158–161].

In coupled biotic–abiotic reactions, intermediates in the nitrification process, such as NH_2OH, NO and NO_2^-, chemically react with oxidizing or reducing substances to form N_2O (hybrid N_2O formation). There are two main pathways for coupled biotic–abiotic processes: chemical denitrification (also known as chemodenitrification) and NH_2OH conversion[156,157,162]. In chemodenitrification, transition metal elements, such as iron and copper, undergo electron transfer reactions and act as electron donors to reduce NO_3^- to produce NO and N_2O under hypoxic conditions[163]. An environment rich in Fe(II) ions or NO_3^- increased the production of N_2O production via chemodenitrification by up to 28% (ref. 155). NH_2OH conversion produces N_2O through two pathways. The first is aerobic hybrid N_2O formation, in which the nitrogen atoms come from NH_2OH and NO_2^- in a 1:1 ratio[162]. The second is the direct oxidation of NH_2OH to N_2O by oxidants (for example Fe^{3+} and MnO_2)[157]. The contributions of these two pathways to N_2O production have not been accurately quantified.

Uncertainties in sources and processes of N_2O production

Increasing atmospheric N_2O has been linked to environmental nitrogen loading; however, the diversity of N_2O producing pathways makes it difficult to predict emissions. There are several key questions and sources of uncertainty regarding the mechanisms and processes of N_2O production that need to be considered.

First, it is important to know how much abiotic processes contribute to N_2O production. The contribution of abiotic N_2O production in forest, grassland, desert and aquatic systems is rarely reported, whereas the process seems to be more often reported in agricultural soils[162,164]. Abiotic processes contribute up to 28% of total N_2O emissions from paddy soils[165,166]. Sterilized soils, in which only abiotic N_2O production can occur, were found to produce 31–75% of the total N_2O produced in non-treated agricultural soils[167]. However, tests on sterilized soils have limitations; for example, sterilization can alter the soil properties, or it can be incomplete, allowing some biotic processes still to occur[168,169]. Despite the limited number of investigations, it is generally assumed that biotic processes dominate over abiotic contributions in agricultural soils[135,170]. A key biogeochemical role of chemodenitrification, which can use microbially derived NH_2OH, NO, NO_2^- or NO_3^-, seems to be the catalytic enhancement of coupled biotic–abiotic N_2O production pathways[171].

Second, there is uncertainty regarding whether NH_4^+-derived N_2O production or NO_3^--derived N_2O production makes the largest contribution to N_2O emissions. Denitrification is a key pathway for N_2O production in natural and engineered ecosystems[9,172]. However, NH_4^+-driven N_2O production has emerged as a major contributor, accounting for up to 60–90% of total emissions in some intensively fertilized agricultural soils[144,173]. Substantial discrepancies exist between estimates of NH_4^+-derived and NO_3^--derived N_2O production owing to the strong heterogeneity of soil and sediment and inconsistencies in research methods; further research is needed to attribute N_2O emissions to different processes in various ecosystems. Dual ^{15}N–^{18}O isotopic tracing suggests that NH_2OH oxidation contributes ~15–40% of marine N_2O emissions[132], whereas ND contributes to ~25–70% in freshwater sediments[103,132] and agricultural soils[144]. NH_4^+-derived pathways might dominate oxic environments, whereas NO_3^--derived pathways prevail in anoxic and hypoxic environments. However, in well-oxygenated coastal waters, denitrification still is the primary N_2O source[141]. Given the contrasting effects of oxygen availability on N_2O production pathways in different environments, no consensus has emerged regarding the conditions under which NH_3 oxidation or denitrification processes dominate N_2O production.

Third, it is crucial to identify the key parameters that influence N_2O production. There are various factors that influence N_2O production, including the substrate, pH, moisture content, oxygen levels and NH_3 levels. For instance, the production of N_2O via the NH_3 oxidation pathways increases as O_2 concentrations decrease, accounting for over 50% of total N_2O produced at low O_2 concentrations (0.5–3%) in soils[144]. In marine environments and rivers, NH_3 and O_2 are the primary sources of nitrogen and oxygen atoms, respectively for N_2O production[141].

Overall, it is important to address these uncertainties regarding N_2O sources, processes, and their responses to environmental change and human activities. Dual ^{15}N–^{18}O isotopic tracing is relatively effective for distinguishing N_2O from nitrification and denitrification, because molecular biology methods alone cannot distinguish the source of oxygen atoms in N_2O. However, research is limited to laboratory investigations, focusing on the mechanism and potential process rate, and neglecting environmental drivers on in situ flux magnitudes. Moreover, in natural environments, biotic and abiotic N_2O production co-occur, and their relative contributions are modulated by environmental factors — particularly anthropogenic warming.

Anaerobic oxidation of NH_4^+ for nitrogen removal

Nitrogen transformations typically involve distinct oxidation and reduction processes. At aerobic–anaerobic interfaces, however, these reactions can co-occur. For example, the anaerobic oxidation of ammonium (anammox), first identified in bioreactors[174] and marine sediments[175], couples NH_4^+ oxidation with NO_2^- reduction, producing N_2 without generating N_2O (refs. 176,177). This pathway is mediated by bacteria within the order *Brocadiales*, which is part of the phylum *Planctomycetes*[178]. Mechanistic investigations across bench-scale, pilot-scale and full-scale systems have consistently demonstrated the ability of anammox to reduce N_2O emissions from the NH_4^+-derived and NO_3^--derived pathways[9,10,179–182]. Initial work has explored the use of anammox for nitrogen removal in sewage and industrial wastewater treatment[178,180]. However, further work is needed to realize its potential for nitrogen removal and N_2O mitigation in natural water bodies.

N_2O mitigation strategies

Mitigating N_2O emissions should involve two aspects: reducing the production of N_2O and increasing its consumption. This section outlines strategies to address N_2O emissions from hotspots (such as cropland soils, small streams and lakes) and rapidly growing sectors (such as aquaculture and waste and sewage treatment).

Mitigating N_2O from agricultural soils

Reducing N_2O production. Agricultural policies and management practices are critical for mitigating N_2O emissions[183] because agriculture is responsible for over 50% of global anthropogenic N_2O emissions, mainly from anthropogenic nitrogen inputs. To effectively mitigate agricultural N_2O emissions, it is essential to moderate the environmental factors that influence N_2O production (for example soil nitrogen and carbon availability, pH, temperature, oxygen supply and water content). Precise nitrogen management strategies can align nitrogen supply with crop nitrogen demand, reducing surplus nitrogen in soil and limiting microbial access to reactive nitrogen species[184]. Remote sensing and AI can provide high-resolution spatiotemporal imagery of N_2O fluxes and improve the predictability of soil N_2O emissions. Such real-time monitoring of soil nitrogen levels could strengthen the applicability of agricultural management strategies and help to optimize nitrogen use efficiency[185].

To achieve sustainable agriculture, measures must not only reduce N_2O emissions but also increase crop yields. Innovative concepts and strategies, such as intercropping and integrated farming systems with diversified crop rotations, have emerged as a means to balance agricultural production (for example grain yield, nitrogen use efficiency and farmer income) with environmental sustainability (for example greenhouse gas emissions, air quality and soil health). These strategies promote rhizosphere processes to aid water and nutrient sharing and maintain the stability of soil chemical and biological properties[186–190].

The implementation of N_2O mitigation strategies is constrained by economic and social barriers, such as a lack of incentives, insufficient nitrogen management knowledge, and policy deficits[191]. One effective policy to encourage farmers to adopt advanced nitrogen management practices is to provide national subsidies to farmers, which can offset the necessary investments in money, time and labour. Additionally, integrating N_2O mitigation into cap-and-trade systems (for example, carbon market and nitrogen credit systems) and allowing farmers to benefit from adopting these practices should be promoted[191,192].

Another effective method for controlling NH_4^+-derived N_2O production is the use of soil additives, such as commercial nitrification inhibitors. These additives generally reduce N_2O production by delaying bacterial oxidation of NH_4^+ to NO_2^- by inhibiting NH_3-oxidizing bacteria activity[193]. Although nitrification inhibitors might increase NH_3 emissions[194], the IPCC has recommended their use to mitigate N_2O emissions[195]. When used in combination with deep fertilization or urease inhibitors[194], nitrification inhibitors could jointly mitigate N_2O and NH_3 emissions in global cropland soils[196]. Many compounds have nitrification inhibitory properties; however, only a few have been commercially produced, including dicyandiamide and nitrapyrin ($ClC_5H_3NCCl_3$)[194]. The use of chemical nitrogen inhibitors such as dicyandiamide can potentially reduce N_2O emissions from urea fertilizers or animal excreta returns in grazed pastures by about 50–70% (ref. 196). This method also reduces NO_3^- leaching, decreasing NO_3^- pollution in groundwater and mitigating the indirect release of N_2O from surface water caused by exchange between surface water and groundwater[196]. The use of chemical nitrogen inhibitors has been widely demonstrated to be viable, and the associated costs are partially or completely offset by production increases caused by the conservation of nitrogen. However, the environmental implications of these inhibitors have not been thoroughly investigated.

It is important to maintain global grain yields; therefore, using crop species that naturally inhibit nitrification would be a practical and feasible strategy for minimizing N_2O emissions. Biological nitrification inhibition (BNI) is hypothesized to be an adaptation mechanism of plants to conserve nitrogen and increase nitrogen use efficiency in nitrogen-limited ecosystems[197]. Plant root exudates from diverse pasture grasses, cereal crops and legume crops containing biological molecules with nitrification inhibition potential have been shown to mitigate N_2O emissions, avoiding the adverse effects of using chemical additives on soil systems[197]. Selecting or breeding (either using conventional or gene editing methods) plant varieties with increased BNI capacity, incorporating BNI crop varieties with non-BNI crops, or using plant growth regulators that stimulate BNI activity could help to reduce N_2O emissions. However, further research is needed to verify the efficacy and viability of using BNIs in agricultural production systems.

Augmenting N_2O consumption. Bioaugmentation refers to the introduction of microorganisms into a natural or engineered environment to increase the rate of a specific process, in this case the conversion of N_2O to N_2 (refs. 198,199). Various microbial nitrogen transformations produce N_2O (Fig. 3), but the only widely recognized biological pathway for its removal is through nosZ, a periplasmic, copper-containing enzyme that catalyses the conversion of N_2O to environmentally benign N_2. Historically, N_2O consumption was thought to be primarily associated with denitrifying bacteria and archaea that reduce NO_3^- and/or NO_2^- to N_2O, and subsequently to N_2 (refs. 200,201). These organisms were believed to be facultative anaerobes, preferring oxygen over NO_x (that is, NO_3^-, NO_2^-, NO and N_2O) respiration.

N_2O-reducing bacteria containing clade I *nosZ* genes, such as *Bradyrhizobium japonicum* and *Bradyrhizobium diazoefficiens*, can be applied as microbial inoculants in soils to promote the reduction of N_2O to N_2 (refs. 202–204). *Bradyrhizobium* spp. have the complete genetic machinery (denitrification genes) to convert NO_3^- to N_2 with transient formation of N_2O. However, the rate of N_2O reduction by *Bradyrhizobium* spp. exceeds the rate at which they and other co-localized N_2O-generating microorganisms produce N_2O. This difference could explain the reduction of N_2O emissions achieved by bioaugmentation of *Bradyrhizobium* spp., assuming that the rate of N_2O exchange between the soil and atmosphere remains constant[202–204]. In field trials, combining engineered *Bradyrhizobium* spp. with high N_2O-reducing activities (Nos^{++}) and symbiotic crop plants (for example soybeans) led to higher colonization by Nos^{++} type *Bradyrhizobium* spp. and reduced N_2O emissions[205].

N_2O-reducing bacteria carrying an 'atypical' *nos*Z (clade II *nos*Z)[206,207], which was discovered through metagenomics, provide further opportunities for bioaugmentation. First, many of these bacteria lack denitrification genes and do not produce N_2O (ref. 208). Second, clade II N_2O-reducing bacteria have a higher affinity for N_2O than clade I N_2O-reducing bacteria[209,210]. N_2O-respiring bacteria, *Cloacibacterium* sp. CB-01, strengthen *NosZ* activity in soils, reducing N_2O emissions derived from organic waste and soil by 50–95% (ref. 211). If bioaugmentation of *Cloacibacterium* sp. CB-01 is scaled up to the European level, national anthropogenic N_2O emissions could be reduced by 5–20%. However, microbiome engineering to strengthen the N_2O reduction capacity of soils is challenging, because inoculants have to overcome various stressors[199]. For example, acidic pH can inhibit microbial N_2O consumption[121], and most N_2O-reducing microorganisms thrive in circumneutral pH environments.

A comprehensive understanding of the physiology and ecology of N_2O-reducing microorganisms is required to develop environment-specific bioaugmentation approaches to mitigate N_2O emissions. Physiological and omic investigations continue to expand the known phylogenetic diversity of N_2O-reducing microorganisms[212,213] and show that the current understanding of the biological N_2O sink remains incomplete. For example, 41.73% of bacterial phylogenetic diversity and 36.39% archaeal phylogenetic diversity lacks any genomic representation[214], indicating that there could be taxa capable of N_2O reduction that have not yet been identified.

N_2O mitigation in terrestrial water bodies

Reducing emissions from sewage treatment plants. The anammox process can remove nitrogen while greatly reducing the need for external organic electron donors (such as methanol) and aeration energy in biological wastewater treatment[215,216]. Although N_2O emissions from wastewater treatment plants can be mitigated by adjusting environmental conditions such as dissolved oxygen levels and implementing post-anoxic denitrification stages[217,218], the rapid rise in N_2O release has not been curbed. N_2O emissions from wastewater treatment could potentially be reduced by approximately 85% by transitioning from conventional nitrification–denitrification to partial nitritation–anammox processes[64], which minimizes NO_2^- accumulation – a key precursor to N_2O formation[64,172]. Additionally, the partial nitritation–anammox process cuts aeration energy by oxidizing only part of the NH_4^+ to NO_2^- and nearly eliminates the need for external organic electron donors, which are required in conventional DD, thereby enhancing cost-effectiveness and operational feasibility in mainstream wastewater treatment. A one-stage reactor integrating partial nitritation in suspended sludge with anammox in a fixed biofilm was implemented for mainstream nitrogen removal at the Fangzhuang wastewater treatment plant in China. This full-scale demonstration of the practical application of anammox technology in municipal sewage treatment validates the effectiveness of this approach for mitigating N_2O emissions in wastewater treatment engineering.

Geoengineering solutions for inland waters. Biogeoengineering is a low-cost, process-controlled technology that uses naturally occurring microorganisms to reduce N_2O emissions and integrates biotechnology and earth science engineering. Anammox bacteria are widely distributed and are particularly prevalent in aquatic ecosystems[58,180,219], contributing to nitrogen loss[181,220]. Anammox hotspots have been found at diverse oxic–anoxic interfaces in terrestrial systems[9,221,222]. Therefore, a nature-based geoengineering approach has been developed to remove NH_4^+, improve water quality and reduce N_2O emissions by means of multiple oxic–anoxic interfaces[179,223], including soil–water interfaces (riparian zone), soil–root interfaces (rhizosphere) and soil–groundwater ecotones (groundwater system) (Fig. 4).

Since beginning operation in 2010, anammox-hotspot geoengineering areas in Jiaxing city, Zhejiang Province, have provided clean drinking water, strengthened CO_2 fixation, reduced N_2O emissions and improved long-term environmental quality. Specifically, the anammox hotspots achieve 43.0% more nitrogen loss, 44.4% more NH_4^+ removal and 27.1% smaller N_2O fluxes than non-hotspot zones[10]. Economic analysis indicates that, despite requiring initial investments in construction, wetland maintenance and reed harvesting, the project reduces water treatment costs, generates revenue through reed sales, and increases property values compared with nearby urban areas. Over a 3-year average, the cost-effectiveness ratio, incorporating social, economic and environmental benefits, is estimated at −19.45 USD m^{-3} (ref. 223). Subsequently, five anammox-hotspot biogeoengineering areas were established in the Jiaxing region of China[223]. This strategy can also be applied to other water bodies with different nitrogen loadings or salinity levels, such as aquaculture ponds, estuaries and coastal vegetation, because anammox bacteria have a high substrate processing ability and are highly adaptable to salinity[224,225].

Biological N_2O fixation to increase N_2O consumption. Directly fixing N_2O from the atmosphere (without its being first reduced to N_2 through denitrification), akin to canonical biological N_2 fixation, is potentially the most straightforward approach to reduce atmospheric N_2O levels. There have been multiple reports of N_2O undersaturation in well-oxygenated freshwater and surface ocean waters that suggested a sink for N_2O, and although the true mechanism was uncharacterized, it was typically ascribed to canonical denitrification[226]. In 2015, evidence for biological N_2O fixation providing a sink for N_2O was reported in the

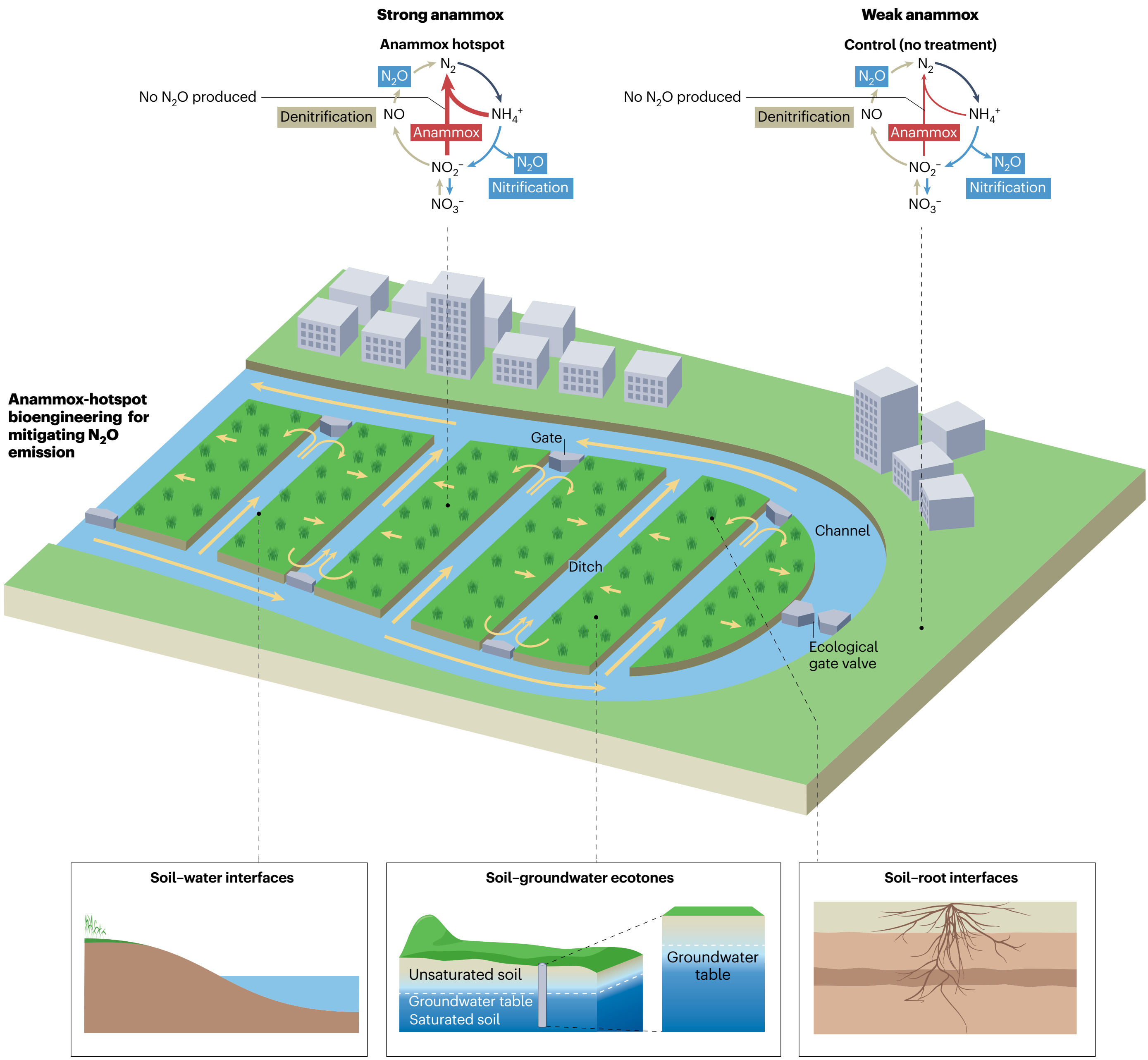


Fig. 4 | Anammox-hotspot biogeoengineering in China. An example of an anammox-hotspot area – the wrinkled Earth Belt, with yellow arrows indicating water flow directions. This biogeoengineering approach increases the number of oxic–anoxic interfaces, including soil–water interfaces (riparian zone), soil–root interfaces (rhizosphere) and soil–groundwater ecotones. Oxidized nitrogen and reduced nitrogen coexist at these interfaces, and they therefore support strong anammox, which does not produce N_2O (top left inset). Conversely, nitrogen cycling in untreated regions is dominated by nitrification and denitrification (top right inset), which produce N_2O. The anammox-hotspot biogeoengineering areas in China form a nature-based geoengineering approach with multiple oxic–anoxic interfaces that mitigate N_2O emissions.

surface waters of the Eastern Tropical South Pacific[227], although the precise mechanism remained uncertain. Subsequent comparison of rates of N_2O and N_2 fixation in freshwater[226] ecosystems confirmed direct biological N_2O fixation as a real alternative to the widely recognized denitrification process as a sink for N_2O (refs. 226,228).

Understanding of the regulation or capacity of biological N_2O fixation is, however, currently insufficient to judge its mitigation potential, and it also remains unclear how widespread N_2O fixation is in the environment[229]. N_2O fixation in freshwater ecosystems seems to be inhibited by inorganic nitrogen, even at very low concentrations

(concentrations of NO_3^- and NH_3 below 1 μM), and seems to be favoured in the cold[226]. Therefore, the application of this process for N_2O mitigation could be limited because high N_2O concentrations are often linked to warmer, highly populated regions with large amounts of inorganic nitrogen pollution[226].

Mitigation of industrial N_2O emissions

Low-cost thermal destruction, catalytic destruction, catalytic reduction and adsorption technologies can mitigate industrial N_2O emissions from fossil fuel combustion and the production of adipic acid and nitric acid[97] (Table 2). Selective catalytic reduction reduces N_2O to N_2 and water, removing up to 80% of N_2O emissions. Catalytic destruction converts N_2O into N_2 and O_2 and is currently being used to mitigate emissions from adipic acid productions[97]. Moreover, the use of photocatalytic and electrocatalytic techniques to reduce N_2O has the advantage of a low carbon footprint and no additional pollution, unlike wet processes which produce secondary pollution such as sulfur dioxide[158,230]. Additionally, emerging techniques have enabled the extensive and effective use of N_2O as a resource for chemical synthesis[231,232]. For example, iron-exchanged zeolites can facilitate the catalytic decomposition of N_2O, while producing value-added chemicals such as adipic acid (a raw material for nylon production) at low temperatures[231]. The choice of technology depends on the industrial process and emission requirements.

Summary and future perspectives

Hotspot N_2O emissions arise from sources such as agricultural soils, tropical forests and thawing permafrost, and hot moments from freeze–thaw cycles, wetting of dry soils and fertilizer application. However, the contributions of specific microbial processes remain overlooked, and some regions are under-studied, leading to uncertainties in N_2O budgets, particularly from natural ecosystems (such as tropical forests) and emerging anthropogenic sources (such as aquaculture and wastewater treatment). N_2O is produced through NH_4^+-derived, NO_3^--derived and coupled biotic–abiotic pathways. These pathways are interlinked and have distinct roles in different ecosystems. However, the contribution of these different mechanisms to N_2O emissions has not yet been distinguished by the IPCC. There is increasing evidence that NH_3 oxidation supplies NO_3^- for denitrification to produce N_2O, and in many ecosystems NH_3 oxidation emits more N_2O than denitrification. As NH_3 fertilizer usage continues to increase, NH_3 oxidation will become an increasingly important source of atmospheric N_2O. Management strategies and practical applications of bioaugmentation and biogeoengineering are being developed to reduce N_2O production and increase N_2O consumption.

The lack of in situ field research, including flux and atmospheric concentration measurements, can lead to discrepancies between flux and process-based biogeochemical or atmospheric inversion model predictions. For example, in tropical forest in the Congo Basin, models underestimated fluxes by ~50% owing to unaccounted pulse emissions during rapid soil rewetting after dry spells. Although isotope tracing and molecular biology methods exist for measuring nitrogen-conversion processes, monitoring N_2O sources at high spatial and temporal resolution across regional to global scales remains challenging, primarily owing to sparse atmospheric observation networks. Consequently, key hotspots and hot moments are overlooked in global estimates, especially in inaccessible regions (such as the Qinghai–Tibet Plateau and Siberian permafrost). Airborne measurements offer a solution by enabling regional-scale flux quantification via eddy covariance or concentration gradients, but they require investment in lightweight sensors, coordinated flight campaigns and robust data assimilation frameworks. Additionally, collaborative research between field and laboratory settings must be strengthened through shared protocols, joint hypothesis-testing campaigns and integrated data platforms to reconcile mechanistic insights with real-world variability.

A comprehensive global N_2O database must be constructed and continuously updated to support the Paris Agreement. This effort involves compiling existing field and remote-sensing data, standardizing metadata (such as environmental drivers and process rates) and creating open-access repositories. The absence of such a database currently leads to large uncertainties in estimates of natural emissions (for example bedrock weathering) and anthropogenic emissions (for example from deforestation, wildfires, and emerging nitrifying and denitrifying niches).

Anthropogenic warming will exacerbate uncertainties in N_2O emission mechanisms. Projected increases in spring–summer and autumn–winter precipitation could stimulate instantaneous N_2O emissions in arid regions. Although dry–wet and freeze–thaw hot moments

Table 2 | Technologies for industrial N_2O mitigation

Technology	Suitable scenarios	Processing efficiency (%)	Advantages	Disadvantages
Thermal destruction[233,234]	Nitric acid production	80–90	No toxic products; efficient use of resources; no secondary pollution; low cost; easy to operate	High energy consumption; high durability requirements
Catalytic destruction[158,159]	Nitric acid and adipic acid production	90–100	Efficient in industrial exhaust environments; high selectivity; high cycling stability	High operation costs; catalyst poisoning
Catalytic reduction[160,230]	Fossil fuel combustion	>90	High activity at low temperature; excellent resistance to sulfur; multipollutant treatment	Ammonia escape not quantified; lacking large-scale application
Adsorption[161]	Exhaust treatment of adipic acid	>60	Low cost; short adsorption time; easy to operate	Complex adsorbent preparation; low efficiency in ambient conditions
Revalorization of N_2O (refs. 231,232)	Production of adipic acid by-products (phenols)	100 at mild conditions	Mild conditions; high selectivity and stability	High costs; not tested for practical applications

dominate annual emissions in some ecosystems, their mechanisms remain unclear. Increasing frequency of these events under warming necessitates urgent study of N_2O sources and cycling in these contexts. Critically, it is not clear how factors such as biogeochemical N_2O production rates and the relative contributions of nitrification and denitrification pathways might shift under warming. To address this uncertainty, integrated experiments combining in situ warming manipulations, controlled lab incubations and isotopic tracing are needed to quantify temperature-sensitive process thresholds. Mitigating N_2O emissions remains challenging where production mechanisms are unresolved.

AI can drive future research by advancing modelling from the molecular to the macroscopic scale. On the molecular scale, AI could be used to decode enzyme structures, reveal quantum states in active sites and identify amino acid combinations critical for N_2O generation. On the genetic scale, AI-integrated multiomics platforms could reconstruct nitrogen metabolic networks and predict gene–environment interactions. On the community scale, AI models could identify optimal microbiome compositions to reduce N_2O emissions while maintaining nitrogen use efficiency. Multiscale models connecting metagenomics, transcriptomics and enzyme kinetics with N_2O fluxes are needed to elucidate how molecular mechanisms influence N_2O emissions. Embedding AI into the sensor–decision–action cycle of agricultural and industrial systems can enable real-time management that balances productivity, efficiency and climate resilience.

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Author contributions
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Competing interests
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